Lesson for the course “Human-Computer Interaction”

The PACT framework & Benchmarking

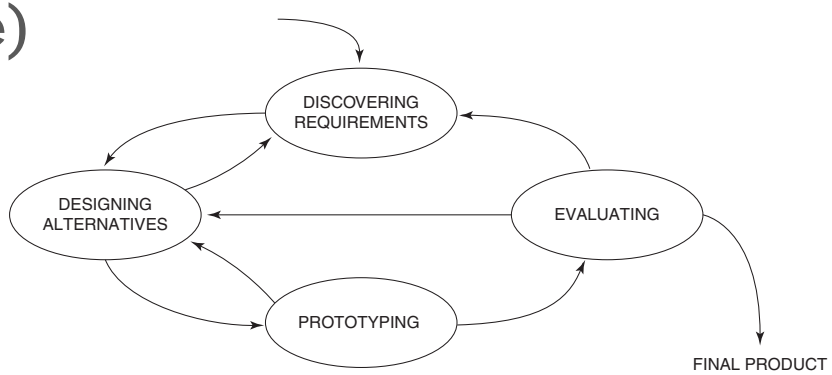
Prof. Luca Turchet



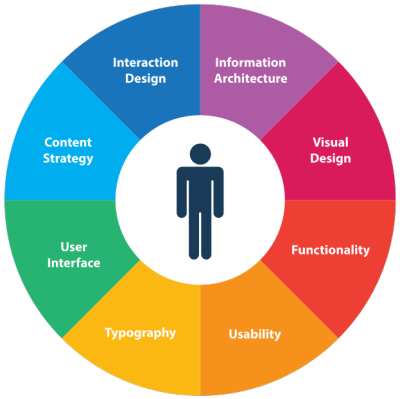
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Recap

- Interaction design (IxD) is a process (4 phases), a creative activity, and is based on **iterations** (designers never get it right first time)



- IxD it is underpinned by the philosophy of **user-centered design**
  - the real users and their goals, not just technology, are the driving force behind product development



- Goal of IxD: to develop **usable** products as well as an **enjoyable user experience**

Recap

- To design for a good experience, **it is important not to jump into decisions from the start**
  - Take a time to ask questions, find the answers, create user stories and explore the web for solutions to similar problems

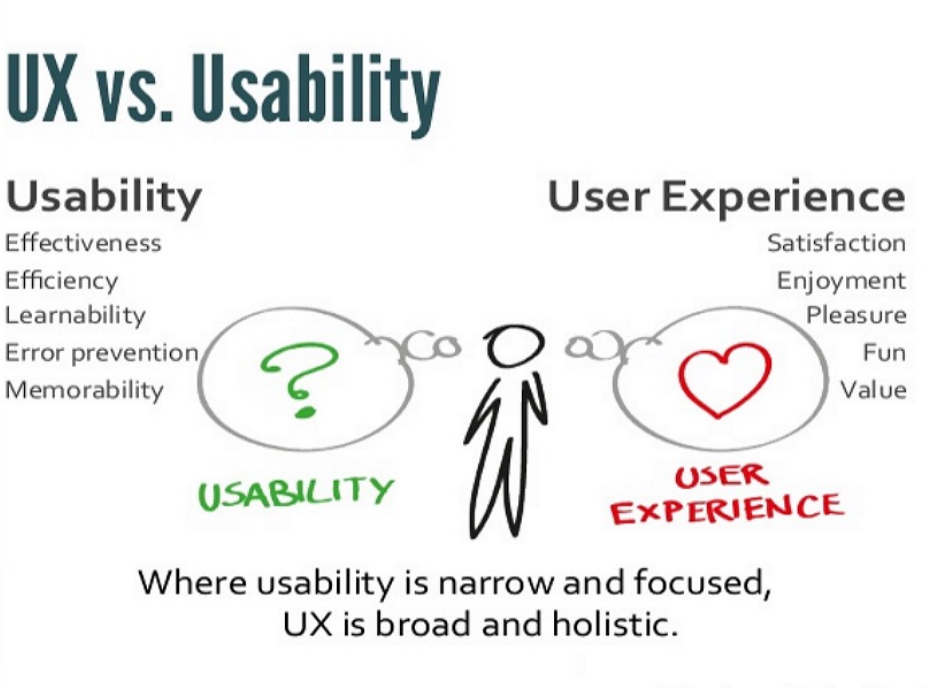
- In the words of Don Norman, *"a brilliant solution to the wrong problem can be worse than no solution at all"*

- **Good designers never start by trying to solve the problem given to them: they start by trying to understand what the real issues are**



Recap

- Usability is a quality attribute of a product. It is **mainly about the functional part of the product**
  - The usability of a product is a crucial part that shapes its UX
- **User experience** refers to a person's subjective feeling and attitudes about using the product



Recap

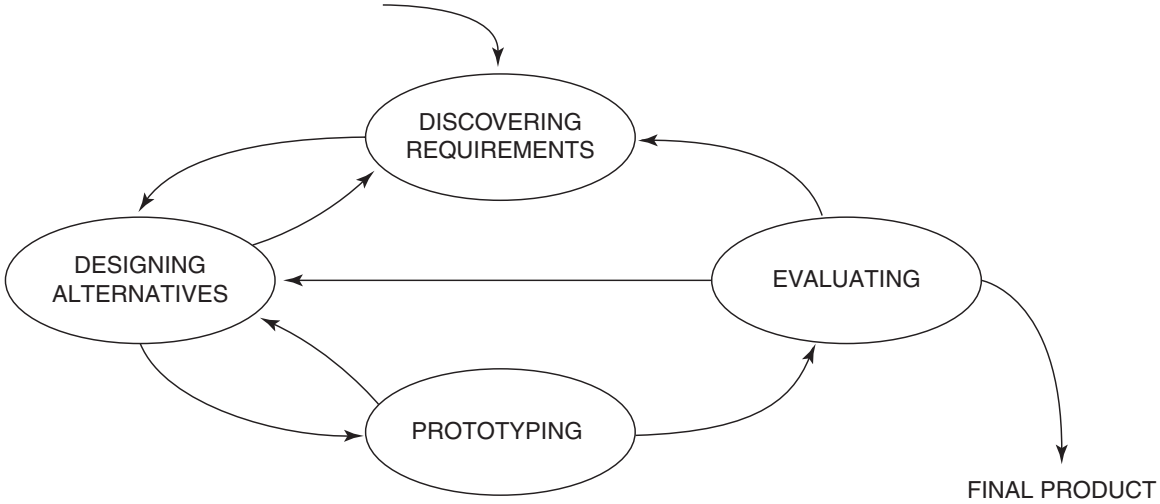
- Usability refers to the **degree** to which products are easy to use, easy to learn, effective, efficient, safe to errors and satisfying to users

- The User Experience is a complex **psychological response** to the interaction with computing systems
  - *"The User Experience encompasses **all aspects** of the end-user's interaction with the company, its services, and its products."* - Nielsen and Norman (2014)

Usability dimensions/goals

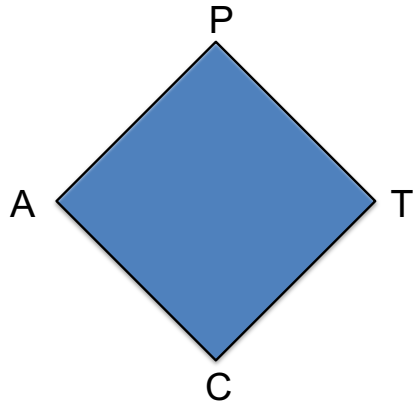
- Usability can be defined in terms of the following dimensions/goals:
  - Effective to use (**effectiveness**)
  - Efficient to use (**efficiency**)
  - Safe to use (**safety**)
  - Having good utility (**utility**)
  - Easy to learn (**learnability**)
  - Easy to remember how to use (**memorability**)
- Usability dimensions are typically put into practice as questions: they allow one to conduct a **usability test**

Exploring the problem space: the PACT Analysis



- Good design takes into account:
  - **Who** the users are – **People**
  - **What activities** are carried out by users – **Activities**
  - **Where** and **when** the interaction takes place – **Context**
  - **What technologies** are used – **Technologies**

The PACT analysis



- Is a User-centric framework for **breaking down a design problem**
- **Take each category** - People, Activities, Context and Technology - **and work through it**
- The analysis helps to focus/orient early design thinking
- Important: **revisit the analysis** (especially at each iteration)
  - As you get deeper into the problem the analysis should change and/or get richer

Exercise: PACT analysis for a product

- Product:
  - A software for tracking and monitoring the mental decline of people at high risk of dementia

- For each category of the PACT framework:
  1. I present the category
  2. You have 3-4 minutes for pen and paper individual reasoning
  3. Short "discussion" on the category

People: who are the users/stakeholders?

- Three categories of users:
  - **Primary**: direct interaction, frequent use
  - **Secondary**: occasional use or via someone else
  - **Tertiary**: affected by its introduction, or will influence its purchase



People: who are the users/stakeholders?

- Who are the primary / secondary / tertiary users of our product?
  - (Product: a software for tracking and monitoring the mental decline of people at high risk of dementia)
- Who are those who:
  - interact directly with the product?
  - manage direct users?
  - receive output from the product?
  - will propose the purchase?
  - make the purchasing decision?
  - use competitor's products?

People: exercise

(Product: a software for tracking and monitoring the mental decline of people at high risk of dementia)

Primary users



Elderly



Doctors

Tertiary users

Secondary users



Care givers



Relatives

Tertiary users

We also need to understand users

- A main reason for having a better understanding of people in the contexts in which they live, work, and learn is that it can **help designers understand how to design** interactive products that provide good user experiences or match a user's needs
- **Understanding individual differences** can also help designers appreciate that **one size does not fit all**
  - What works for one user group may be totally inappropriate for another
  - For example, children have different expectations than adults about how they want to learn or play
- Learning more about people and what they do can also **reveal incorrect assumptions** that designers may have about particular user groups and what they need

People: variability

- Humans vary in many aspects
- **Physically**
  - Age differences, physical abilities
- **Psychologically**
  - Attention, perception, memory
    - They affect the ability to form the right "conceptual model" of the machine
- **Socially** and **Culturally**
  - Level of education
  - Level of familiarity with technology





People: physical variability

- **Motor abilities** may affect the suitability of certain input and output devices



- **Size of hands** and **dexterity** may affect the size and positioning of input buttons

- Consider **height** if designing a physical kiosk



People: physical variability

- **Strength** - a child's toy requires little strength to operate, but greater strength to change batteries

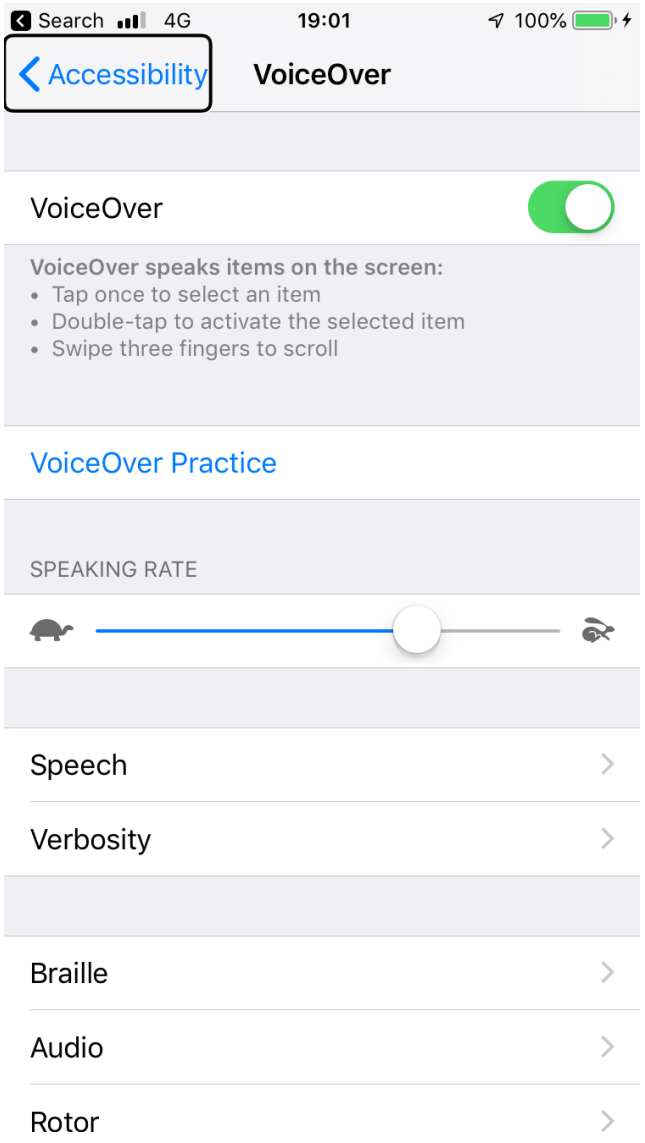


- **Different sensory abilities** (e.g., sight, hearing)



Accessibility and inclusiveness

- **Accessibility:** the extent to which an interactive product is accessible by as many people as possible
  - The focus is on people with disabilities
  - E.g., Apple VoiceOver (for blinds)
- **Inclusiveness:** making products and services accommodate the widest possible number of people
  - E.g., smartphones designed for all and made available to everyone regardless of their disability, education, age, or income



People: Cognitive variability

- Level of **cognitive development** due to age (children vs. adults vs. elderly)



- Cognitive **impairments**

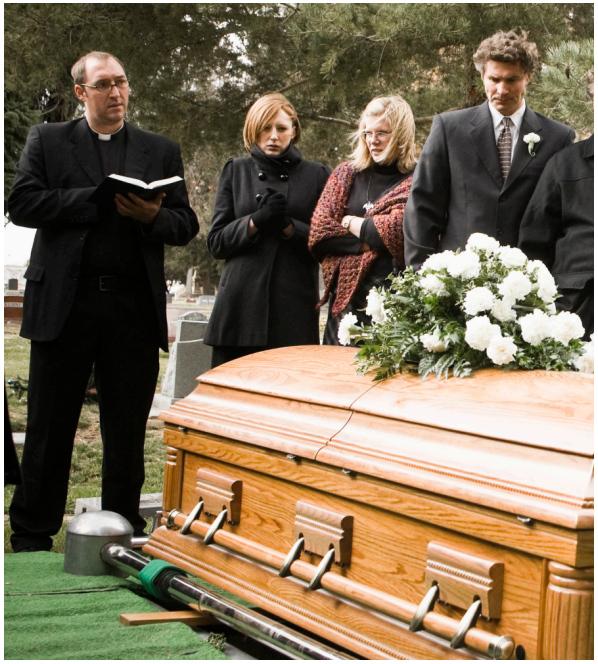


- Momentary deficit: **fatigue, stress**



People: cultural differences

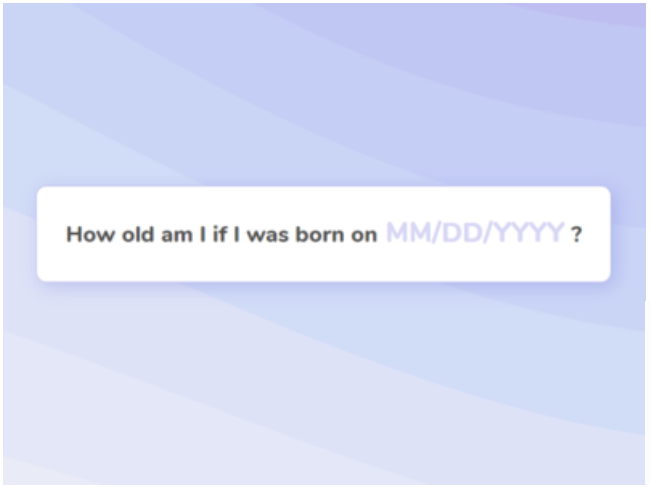
- Being aware of **cultural differences** is also an important concern for interaction design, particularly for products intended for a diverse range of user groups from different countries
- An example: the meaning of colors





Understanding users: cultural differences

- Dates and times used in different countries
  - In the US: month, day, year (05/21/20),
  - In other countries: day, month, year (21/05/20)
- This can cause problems for designers when deciding on the format of online forms, especially if intended for global use
- Solutions: **you don't let the users input the date in a form without any or loose constraint, but you force them to input it in a specific way**





Here the user can still input the wrong date, despite the indication

Activities

- What is the overall purpose of the activity?
  - What has to be satisfied?
  - Hedonic or pragmatic?
- Temporal aspect
  - Regular or infrequent?
  - Time pressure?
  - Continuous or interruptions?
  - Processing time?
- Cooperation
  - One or more actors?

Activities

- Complexity of the activity
  - Well defined or vague?
- Safety critical activity
  - Impact of error (how much?)
- The nature of the content of the activity
  - Type of data to be processed
  - Type of media involved

Activities: exercise

(Product: a software for tracking and monitoring the mental decline of people at high risk of dementia)

- Pragmatic / hedonic?
- Regular / infrequent?
- Time pressure?
- Continuous / interrupted?
- Processing time?
- Cooperative or not?
- Well-defined / vague?
- Safety critical?
- Nature of the content?

Context

- Where/when does the activity/interaction occur?
- The context can be
  - **Physical context**
    - At home, in the office, on the move
    - Noisy, scarce light, bad weather
  - **Social context**
    - Individual activity vs. group activity (e.g., training with the Wii)
    - Computer-mediated social activity (e.g., co-writing a Google doc)
    - Social norms
  - **Psychological context**
    - Motivation, attitudes
    - Stress
    - Emotions

Context: exercise

(Product: a software for tracking and monitoring the mental decline of people at high risk of dementia)

- What is the physical / social / psychological context of the use of our product?

Context: exercise

- Physical: at home / elderly-care center
- Social: individual
- Psychological: stressful?

Technology

- Input
  - Getting data in, security issues
- Output
  - Video vs pictures; audio speech vs screen
- Communication
  - Between people, between devices, web-based
- Content
  - What content to show?

Technology: exercise

(Product: a software for tracking and monitoring the mental decline of people at high risk of dementia)

- What technologies are currently used by those users?
- What type of technology can be envisioned for our product?
  - Input
  - Output
  - Communication
  - Content

Technology: exercise

- Computer and/or mobile
- Multi-media support
- Internet


BENCHMARKING

Benchmarking

- Method created and mostly used in the **business** sector
- Aimed to identify the best products / practices in a certain industry (or in similar ones)
- Benchmarks are **reference points** that you use to **compare** something
  - e.g., your performance against the performance of others
- By comparing the products / practices identified, it is possible to understand
  - What market needs they cover
  - How these products differentiate one from the other
  - How well they perform

Goals of benchmarking

- **To have a general overview** of the current systems that are related to the problem you are designing for
  - This may help to identify a **gap** to work on with your project
- **To get inspiration for the design**
  - It works as a **design library**, i.e., an organized collection of best practices and examples to take inspiration from

How to make a benchmarking

1. **Step 1:** collect from the web **examples** of systems related to your problem
   - E.g., systems for fostering sustainable mobility

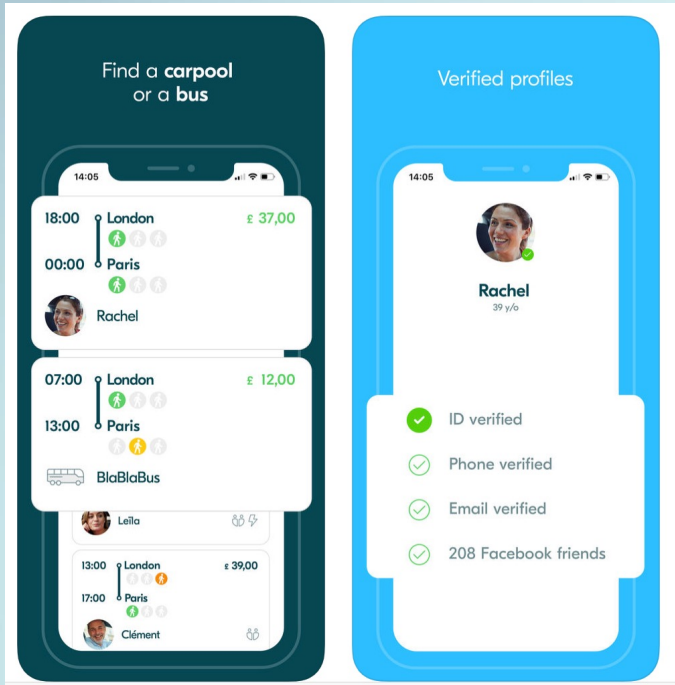
- Suggestions:
  - **Focus on a domain**
  - **Use several keywords**
  - **Local, national, international examples**
  - **Don't look only at apps and websites**
    - E.g., check tangible interfaces, wearable devices, games, etc.

How to make a benchmarking

- A possible way to collect and organize the examples could be to **create a presentation** where you take note of:
  - Name of the system
  - A representative image
  - A short description of how the system works
  - URL
  - Keywords that summarize why that example is interesting

Example slide of benchmarking presentation

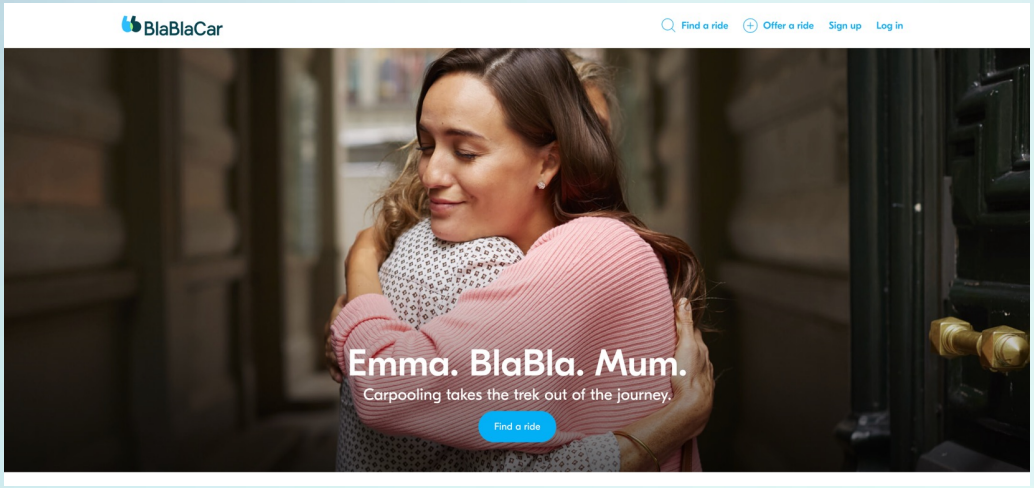
BlaBlaCar



- An online platform for carpooling
- Its website and app connect drivers and passengers available to travel together between cities and share the travel costs

Example slide of design library

BlaBlaCar



- Hero graphics at full screen (emotional impact)
- "Find a ride" directly accessible

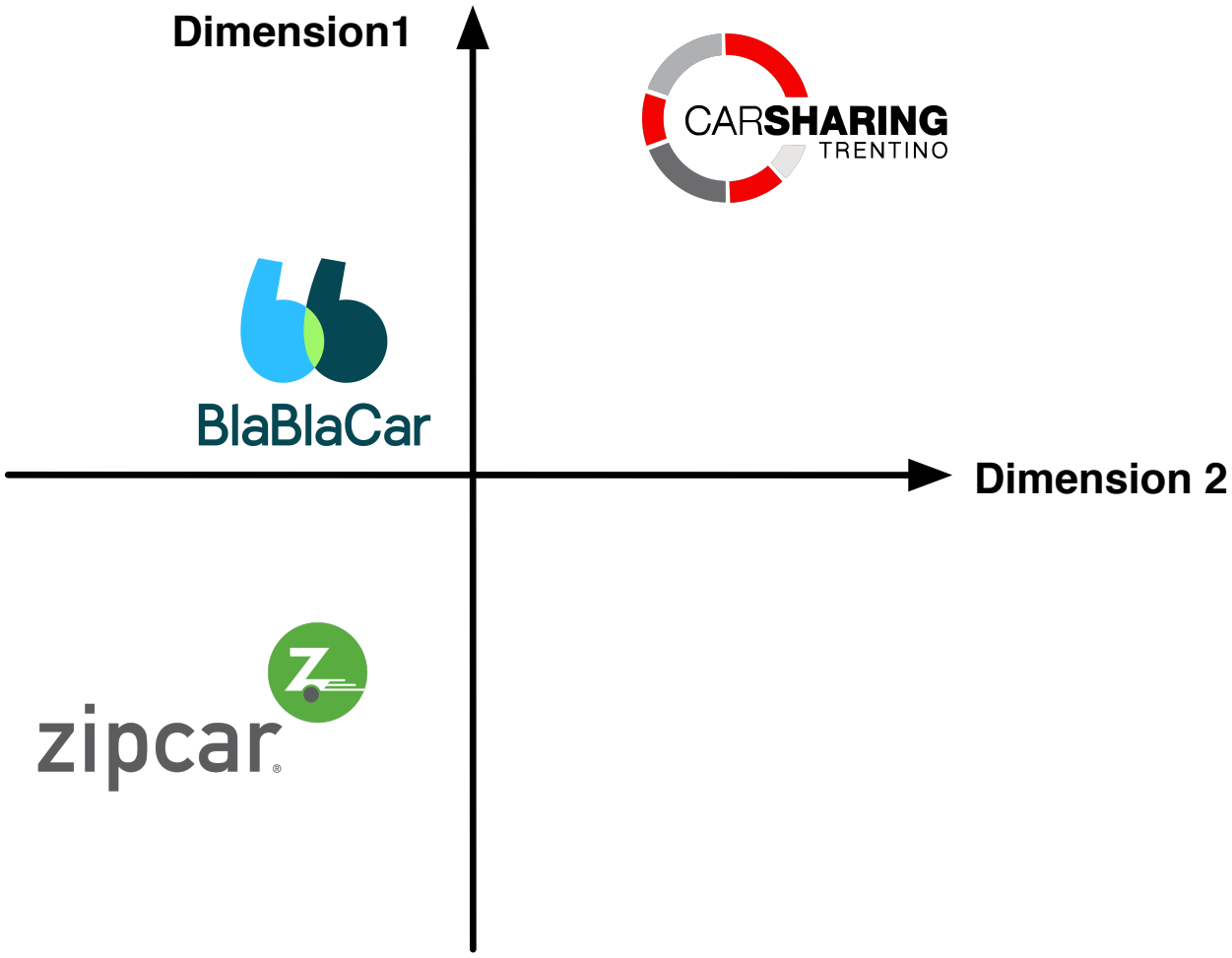
- Note: you can also list bad design examples

How to make a benchmarking

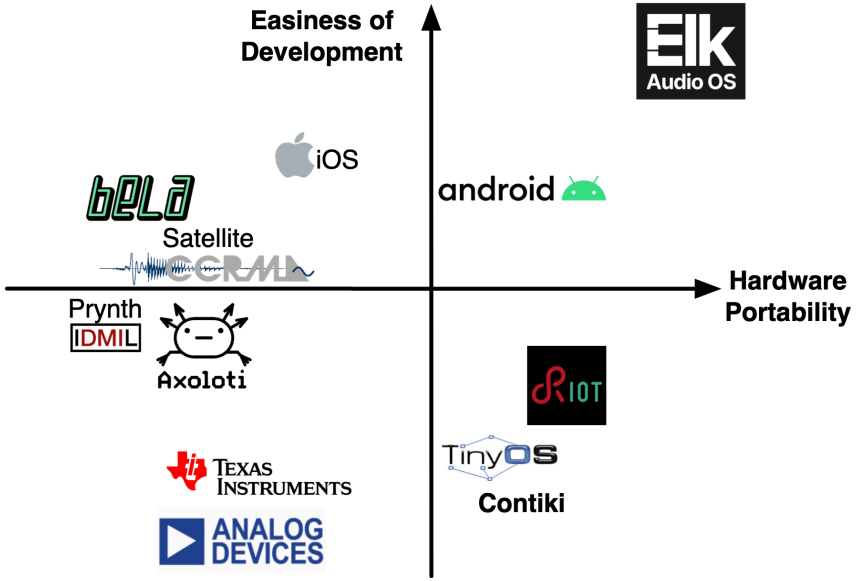
2. **Step2:** classify the examples found **by affinity** (e.g., thematic, target users, PACT, design strategy adopted, etc.)

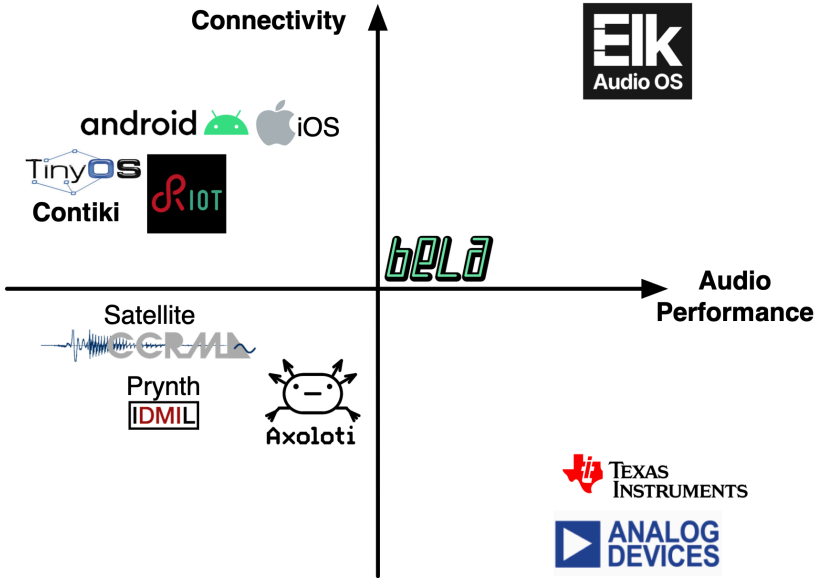
- Tools for visual organization:
  - **Orthogonal axes**
  - **Conceptual map**
    - hand drawn
    - or through software tools, e.g.,
      - https://www.xmind.net/ (not free)
      - https://www.draw.io/ (free)

Benchmarking: classification using orthogonal axes

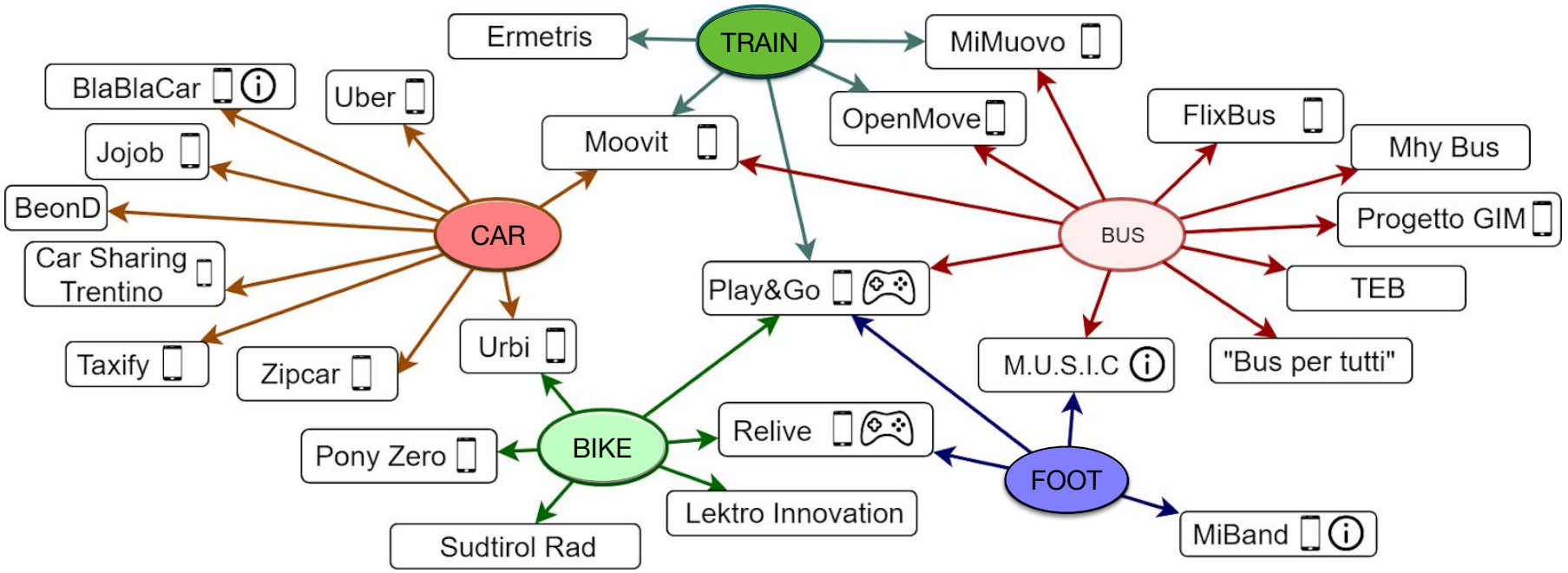


Benchmarking: classification using orthogonal axes





Benchmarking: classification using conceptual maps



Raises awareness

Uses gamification

Uses a technological approach

The words of the expert



Observe, Test, Iterate, and Learn (Don Norman)

9.333 visualizzazioni

280 2 CONDIVIDI SALVA

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There isn't a next time in product development. You must always study to keep up with the product cycle.

https://youtu.be/JgPppwsocRU

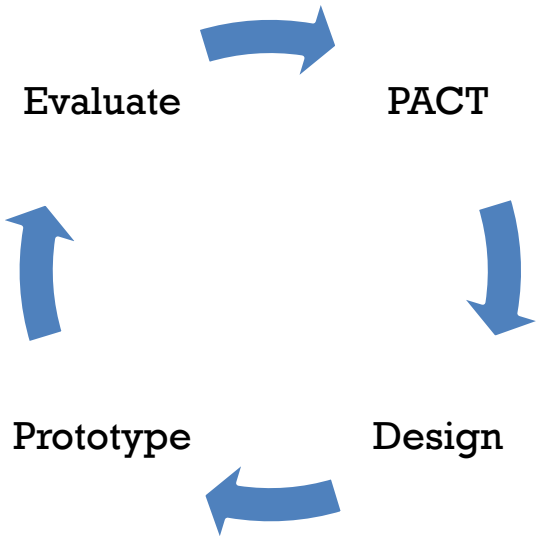
Topics of next lessons

- We will see how to plan and run a successful data gathering program
  - How to plan and run an interview
  - How to design a questionnaire
  - How to plan and carry out an observation
- These techniques will provide us with the data we need to **discover the user requirements**
- In the subsequent lesson we will see how to analyze, interpret and present those data

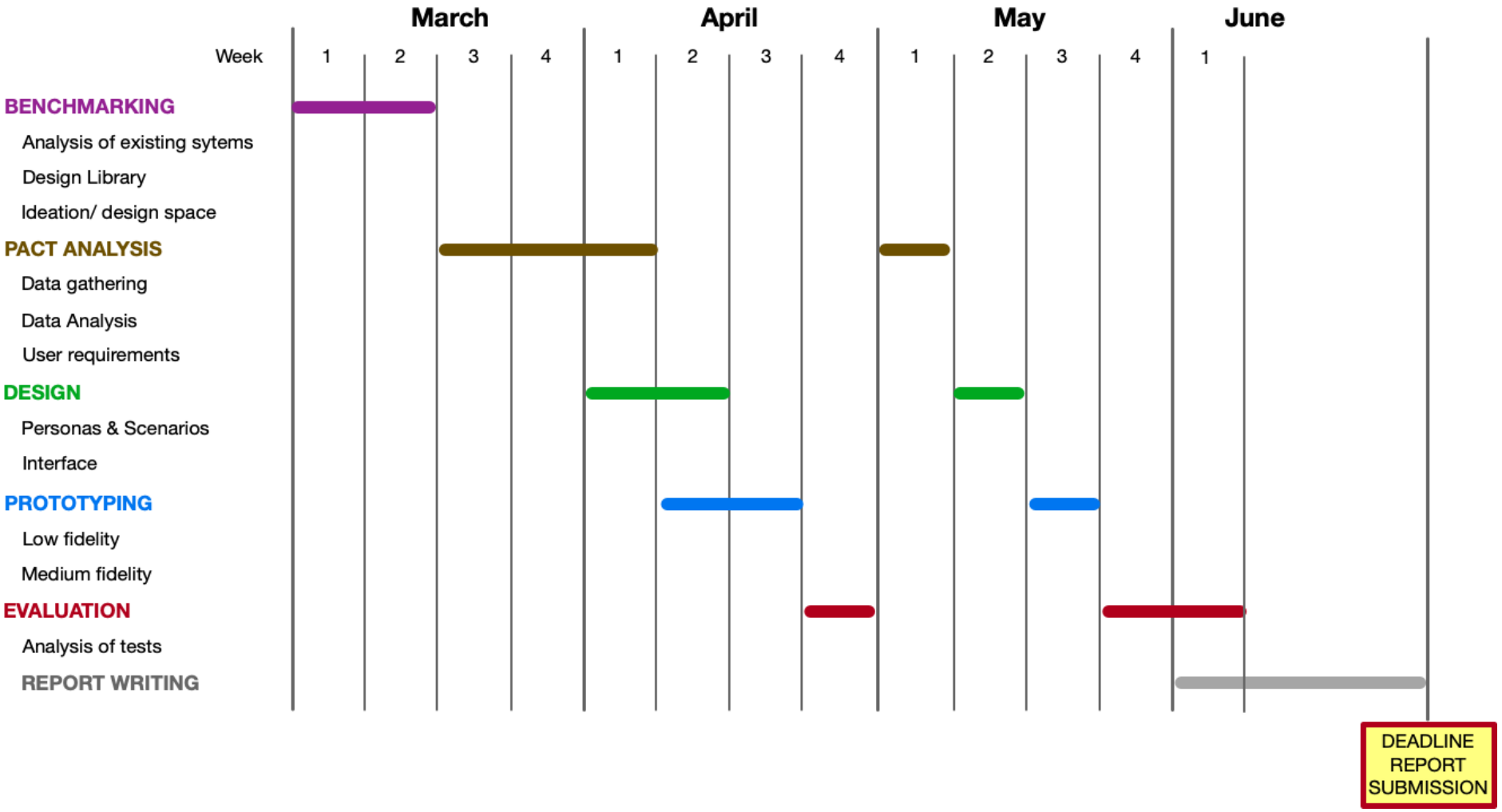
Exam: project work phases

1. Analysis of existing products (benchmarking)
2. Ideation (i.e., identification of the design space)
3. PACT analysis
4. Design
5. Prototyping
6. Evaluation

2 Iterations



GANTT Chart


March
April
May
June
Week
1
2
3
4
1
2
3
4
1
2
3
4
1
BENCHMARKING
Analysis of existing sytems
Design Library
Ideation/ design space
PACT ANALYSIS
Data gathering
Data Analysis
User requirements
DESIGN
Personas & Scenarios
Interface
PROTOTYPING
Low fidelity
Medium fidelity
EVALUATION
Analysis of tests
REPORT WRITING
DEADLINE
REPORT
SUBMISSION

Note

- For your project you must involve participants who
  - Are older than 18 years old
  - Do not belong to a vulnerable population
- In doubt, please ask me for an opinion

Homework

- Choose the topic of your project
- Start the benchmarking phase
- Create the benchmarking presentation
- Compare the identified solutions (e.g., on the orthogonal axes)
- Create the design library
- Start the ideation phase

Bibliography

- Chapter 1 of the book "Interaction Design: beyond human-computer interaction" – 5th Edition, by Sharp, Rogers, and Preece. Wiley, 2019.
- Chapter 2 of the book "Designing Interactive Systems – A comprehensive guide to HCI, UX and interaction design" – 3rd Edition by David Benyon
- https://uxdesign.cc/inclusion-in-design-843f99b5d1c8